

A Hybrid Transformer-Based Sentiment Analysis Model with Emoji-Slang Normalization and Sarcasm Detection

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Abstract- social media has become a major source of real-time opinions, emotions, and public reactions, especially through platforms like X. This is accompanied by a rapid growth in volumes of uninstructed textual data generated by users, creating new opportunities for automated sentiment interpretation. This work outlines a hybrid sentiment analysis framework that merges classic machine learning with modern deep learning for more reliable context-sensitive sentiment classification. This work proposes a system that integrates a transformer-based DistillBERT model for contextual understanding, logistic regression with TF-IDF for lexical precision, and a bidirectional LSTM for sequential dependency learning, all combined through a weighted ensemble approach. An extra preprocessing layer was added to normalize emojis and slang expression, while a sarcasm detection module using RoBERTa adjusts the prediction when irony is detected. This model is trained and evaluated on the Sentiment140 dataset, achieving improved accuracy and robustness compared to typical TF-IDF -based classifiers. These results prove that a hybrid design will provide a greater capacity to handle diverse linguistic expressions, improve performance on noisy data, and give stable predictions across different contexts. The proposed framework has great potential in practical applications such as brand reputation monitoring, social media analytics, and public sentiment tracking. Besides, by incorporating classic interpretability with deep contextual reasoning the system achieves a good balance between transparency and intelligence, thus laying the foundation for future research in emotion-aware and explainable AI- powered sentiment analysis.

Keywords – Sentiment Analysis, Machine learning, Natural Language processing(NLP), Transformer models, DistillBERT, BiLSTM, TF-IDF, Sarcasm detection, Social media analytics

I. INTRODUCTION

Before the widespread availability of advanced Natural Language Processing (NLP) models, organisations and researchers relied on manual processes like surveys, reviews, and customer feedback forms to get an understanding of opinions and market sentiment. With the increased use of the Internet and social media sites, such as Twitter, Facebook, and Instagram, people have increasingly used these platforms to

publish online sentiments and communicate their thoughts and feelings. In just one minute, millions of users post their opinions, experiences, and reactions as short-text messages, often using emojis or abbreviations to express themselves. All of this contributes to significant growth in unstructured data and is one of the reasons that Sentiment Analysis (SA) has become a more significant aspect of subjective information extraction, and the ability to study human emotions in digital communication has greatly increased in importance. Sentiment analysis is different from traditional data analysis methods, which primarily rely on numerical values to gain meaning, by extracting opinion polarity or intensity out of natural language text. Sentiment analysis aims to classify text as positive, negative, or neutral, while attempting advanced approaches to differentiate between feelings such as sarcasm, irony, or mixed sentiments. The last 10 years have seen SA applied to a range of areas, including social media monitoring, financial forecasting, healthcare analytics, disaster analysis, and customer service automation.

The capacity to analyze and distill textual feedback into quantitative measures has led to the increasing use of sentiment analysis as a component of data-driven decision-making in the commercial and research space. While there are now many versions of sentiment analysis software in use, challenges remain to effective use. For example, the textual nature of many social platforms (such as Twitter, etc.) means that the information is often ‘noisy’ and contains slang, abbreviations, emojis, and sarcasm that often breaks apart the standard meaning of words. Many of the traditional machine learning models, such as Support Vector Machines (SVM), Naïve Bayes, and Logistic Regression, based on statistical features such as bag-of-words or TF-IDF, are often unable to deal with these contextual variations in language. The advent of deep learning models, particularly Recurrent Neural Networks (RNN) and Long Short-Term Memory (LSTM) networks, aimed to deal with these limitations by providing models that were able to capture sequential dependencies and contextual dependencies between words. Although these deep learning networks produced improved predictions, they still

faced limitations with ambiguity in context and failed to capture sentiment inversion, particularly sarcastic remarks, and other more sophisticated sentiments. The development of Transformer-based models BERT, RoBERTa, and DistilBERT has greatly improved the performance of NLP applications with their capacity to understand words in a bidirectional context. These models outperform other models in state-of-the-art performance on a number of tasks including question answering, machine translation, and sentiment analysis.

Nevertheless, even the improved architectures equipped for deep contextual modeling (as described in this research) still have limitations handling informal language patterns or emoji-derived sentiments of social media content. To resolve these issues, this research introduces a hybrid transformer-based approach to perform sentiment analysis by benefiting from a traditional and deep learning methods, incorporating the output of contextual representation from DistilBERT, the statistical features from TF-IDF with Logistic Regression, and sequential representation from a Bidirectional LSTM (BiLSTM) network. Furthermore, added to the approach is a sarcasm detection component developed from the RoBERTa model, and an emoji-slang normalization pipeline to improve the interpretation of emotions expressed with emojis. Collectively, the outputs of each of these models are combined using a weighted ensemble approach, meaning that precision, interpretability, and contextualization are automatically weighed and used to classify sentiment outcomes and combine them into an overall outcome.

This Hybrid approach has the three multiplicative goals to:

1. Increase accurate sentiment classification on noisy and informal Twitter data, with multi-model fusion.
2. Interpret emoji and slang expressions that traditional preprocessing often throw away or misunderstand.
3. Flexibly flip sentiment polarity based on detection of sarcasm, and limit false classifications in sarcastic or ironic sentiments.

The research employs the Sentiment140 dataset that consists of 1.6 million labelled tweets, providing a large and heterogeneous textual corpus, with high variability, for training and testing the model. By employing a single architecture that combines multiple learning paradigms, the proposed system achieves improvements in classification performance and shows enhancements in robustness, interpretability, and scalability. The findings highlight the increasing importance of hybrid architectures to address the complexities of human emotions and the limitations of machine interpretation; thus, contributing to the improved design of intelligent, context-aware systems for sentiment analysis in real-life applications.

II. LITERATURE REVIEW

Sentiment Analysis (SA) has progressed remarkably from lexicon-based approaches to advanced hybrid architectures that leverage deep learning and transformer models for contextual understanding. Early works relied on rule-based or dictionary-driven approaches, such as SentiWordNet and AFINN, which assigned polarity scores to words to derive sentiment orientation [1]. Although efficient for structured text, these methods underperformed on unstructured and informal data such as Twitter posts, where sarcasm, emojis, and abbreviations distort lexical meaning [2]. Machine learning (ML)-based models, including Naïve Bayes (NB),

Support Vector Machine (SVM), and Logistic Regression (LR), improved accuracy by learning statistical relationships between tokens, often using Bag-of-Words (BoW) or Term Frequency-Inverse Document Frequency (TF-IDF) representations [3]. Comparative studies, such as those by Rahat et al. [4], showed that SVM outperforms NB in precision and robustness across product-review datasets. However, frequency-based representations failed to model contextual and semantic dependencies between words, especially in short, noisy social texts [5].

Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) emerged, learning hierarchical and sequential dependencies in textual data. Among them, Long Short-Term Memory (LSTM) networks gained prominence for their ability to retain long-term dependencies and mitigate vanishing gradient issues [6]. LSTM-based sentiment classifiers demonstrated superior performance on Twitter datasets compared to classical ML methods by capturing dynamic linguistic patterns and contextual cues [7]. Extensions such as Bidirectional LSTM (BiLSTM) further improved performance by processing text sequences in both forward and backward directions, thereby enhancing contextual understanding [8]. However, such sequential models require large labeled datasets and often lack interpretability or scalability for real-time applications.

Bouassida and Mezali [9] presented a milestone study that proposed hybrid transformer-sequence architectures such as BERT-BiLSTM, RoBERTa-CNN-BiLSTM, and DistilBERT-BiLSTM, evaluated on the Sentiment140 dataset. Their work demonstrated that coupling transformers with recurrent layers improved semantic representation and classification accuracy over standalone architectures. Nonetheless, the study did not incorporate emoji normalization, slang handling, or sarcasm detection, which are vital for analyzing informal Twitter discourse. Similarly, Khan et al. [10] introduced an emoji-fused sentiment model using BERT and classical ML algorithms, highlighting the importance of preserving emoji semantics. Their experiments on airline review datasets showed that encoding emojis instead of removing them significantly enhances classification performance. Yet, their approach lacked sarcasm handling and ensemble integration, limiting its real-world robustness.

III. PROPOSED METHOD

In this project, A Hybrid sentiment analysis system for Twitter data is developed to improve the accuracy and reliability of sentiment classification. The system combines three main models – A transformer model, a logistic regression model, and an LSTM -based sequential model along with a separate sarcasm detection module. Each model focuses on a specific linguistic aspect: the transformer learns contextual meaning, the logistic regression model captures word – level importance, and LSTM model learns sequential patterns. The sarcasm detector corrects cases where the literal meaning of a tweet differs from its actual intent. All components work together through a weighted ensemble mechanism, and the input data is preprocessed through an emoji- and slang-aware pipeline. This design helps the system handle the informal and emotionally diverse nature of social media text.

1. Transformer Expert - DistillBERT:

The first model in the system is DistillBERT, a lightweight version of the BERT transformer model. It serves as the main sentiment classifier and focuses on understanding the context of each sentences rather than individual words. DistillBERT retrains around 97% of BERT's accuracy which suits for large-scale and real-time applications. In this project, the model was fine-tuned on 800,000 tweets from the Sentiment140 dataset, which includes equal numbers of positive and negative samples. Tweets are first tokenized using pre-trained DistillBERT tokenizer and then converted into contextual embeddings using self-attention layers that capture both left and right dependences between words. The output from the final classification layer generates probabilities for positive or negative sentiment. DistillBERT contributes a deep understanding of language semantics, enabling it to detect tone shifts and emotions in informal social media text.

2. Statistical Expert - TF-IDF + Logistic Regression:

The second model is a Logistic Regression(LR) classifier with TF-IDF. This model plays a key role in interpretability and quick prediction. The TF-IDF vectorizer converts each tweet into numerical form, emphasizing words that are more frequent in one sentiment class but rare in another. Logistic Regression is then applied with a regularization parameter ($C = 1.0$) and the liblinear solver. The model which is trained on the subset of 50,000 tweets to maintain computational efficiency. It achieved a validation accuracy of 78.56%, acting as a strong lexical baseline in the ensemble. The Main advantage this component is that it can handle abbreviations, slang, that the Transformer model alone lacks it. It also supports explainability, allowing us to interpret which words have the strongest influence on the sentiment classification.

3. Sequential Expert - LSTM:

The third model focuses on the order and flow of words in each tweet using a long Short-Term Memory network. LSTM's are designed to understand time-based or sequential patterns in text. This model helps the system detect how the meaning of sentences change depending on word order, which is important for cases like negation or sarcasm. The LSTM used in this work consists of two layers, each with 128 hidden units and an 0.5 Dropout rate used to prevent overfitting. An embedding layer of dimension 100 converts words into dense vectors before feeding them into layer of dimension 100 converts words into dense vectors before feeding them into the LSTM. The model was trained on 50,000 tweets for 5 epochs, achieving a validation accuracy of 74.49%. It is particularly useful for identifying complex expressions where the structure of the sentence affects the true meaning. Overall, the LSTM adds sequential understanding that complements the contextual strength of the transformer and lexical precision of the LR model.

4. Sarcasm Detector — RoBERTa-Irony:

To improve robustness against sarcasm and irony, a separate RoBERTa-Irony model was integrated into the system. This module uses a pre-trained transformer model developed by CardiffNLP, specifically trained on sarcastic tweets. The model analyses tweets and produces a sarcasm probability score(p_{sarc}). If the sarcasm score exceeds a defined threshold

(typically 0.9), the sentiment output from the ensemble is corrected by either inverting or reducing its confidence. For example such as “*Love it when my phone dies right before an important call*” might be misclassified as positive with out sarcasm correction, but with this module, it is correctly identified as negative. This module is essential for improving real-world accuracy with sarcastic and emoji detected sentences in the Twitter data.

5. Weighted Ensemble Integration:

The Final sentiment prediction is obtained through a weighted ensemble fusion of the three sentiment models. Where every model will contribute towards the final point according to its weight, depending on its performance and reliability. The formula is given as:

expressed in equation (1):

$$P_{\text{final}} = w_t P_{\text{Transformer}} + w_{lr} P_{\text{LR}} + w_{lstm} P_{\text{LSTM}} \quad (1)$$

Where $w_t = 0.6$, $w_{lr} = 0.2$, and $w_{lstm} = 0.2$

This weighted combination allows the system to balance the transformer's contextual understanding, the logistic regression model's lexical precision, and the LSTM's sequential reasoning. After fusion, the sarcasm adjustment rule is applied if $p_{sarc} > 0.9$:

$$P'_{\text{final}} = (1 - P_{\text{final}}) \text{ or } P'_{\text{final}} = P_{\text{final}} \times (1 - \alpha) \quad (1.1)$$

Where $\alpha = 0.7$. This ensures that sarcastic tweets are reclassified correctly based on contextual cues. The final output includes both sentiment label(positive or negative) and a confidence score.

By combining these models, the proposed hybrid system provides improved accuracy, contextual understanding, and resistance to sarcasm, making it suitable for real-world Twitter sentiment analysis tasks.

IV. DATA COLLECTION AND PREPROCESSING:

The experimental setup employs the Sentiment140 dataset, a sizable and diverse corpus of 1.6 million annotated tweets, with a total sampling record of 800,000 tweets (400 k positive + 400 k negative) selected for training and assessing the models. Each record consists of a short text message and a binary sentiment label derived automatically from user-provided emoticons. This dataset offers a realistic basis for examining informal social-media language that includes emojis, hashtags, abbreviations, and user mentions—all complexity barriers for traditional NLP (natural language processing) pipelines. The dataset was loaded directly from CSV, confirmed for class balance, and partitioned into training, validation, and test sets in a 90 : 10 : 10 ratio. Subsequently, 50 k tweets were sampled for classical and sequential models (TF-IDF + LR and LSTM) to optimize computation, while the complete 800 k corpus was used for fine-tuning the transformer.

1. Normalization of Language and Noise Removal

The preprocessing pipeline initiates by launching an automatic labor to reorder or remove raw tweets for detracting functional non-linguistic noise while maintaining the desired structure of any signal emitted from the tweets sentiment - positive, neutral or negative. The normalization procedure notes and sets all

words to lowercase, removes URLs, usernames and punctuation by repetitive regular expressions, separates all hashtags into meaningful word shaking at the relative time required by the hashtags functional meaning, changes emoji into textual auspice using an emoji lexicon, uses a custom slang dictionary to expand abbreviations, e.g., 'luv' → 'love' and 'gr8' → 'great,' towards linguistic and through line representation. It even identifies polarity carrying stop words, e.g., 'not' and 'never', to keep for the sake of preliminary resetting of polarity. The outcome of this procedure is normalized and sentiment preserved linguistics for hybrid modelling.

2. Tokenization and Vectorization

Tokenization of the cleaned tweets are done according to specifications for the models. The DistilBERT tokenizer converts the text into subword tokens and attention masks, using padding or truncating to ensure all sequences to a standardized 128 tokens long or short. The Logistic Regression classifier and model use a TF-IDF vectorizer with unigrams and bigrams to encode a statistical importance of terms noted in context of the tweets from the tweets, while the LSTM uses a CountVectorizer based on a 10,000-word vocabulary for indexing the tweets into sequences before embedding them. This provides a consistent and fixed structure for input the dimensionality and features for the inputs in all the contextual-, statistical and sequence-based models.

3. Dataset for Hybrid Training

Three datasets were created for using the Sentiment140 dataset: the tensor model used the full 800,000-tweet dataset (learning rate 2×10^{-5} , AdamW optimizer, 3 epochs, and batch size 8) and while the TF-IDF + LR models and LSTM were trained on 50,000 stratified samples each for five epochs each. The validated accuracies are 78.56% for TF-IDF + LR and 74.49% for LSTM. The model checkpoint, vectorizer and vocabulary1 was kept for inference, and the training in creating a hybrid system where training in a modular manner and parallel for hybridization in an ensemble system was successful.

V. SYSTEM ARCHITECTURE AND FLOWCHART:

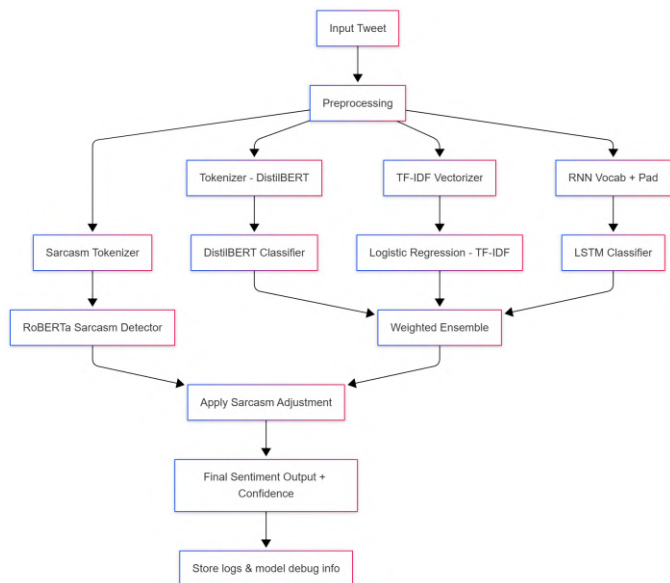


Fig. 1 - Proposed System Architecture

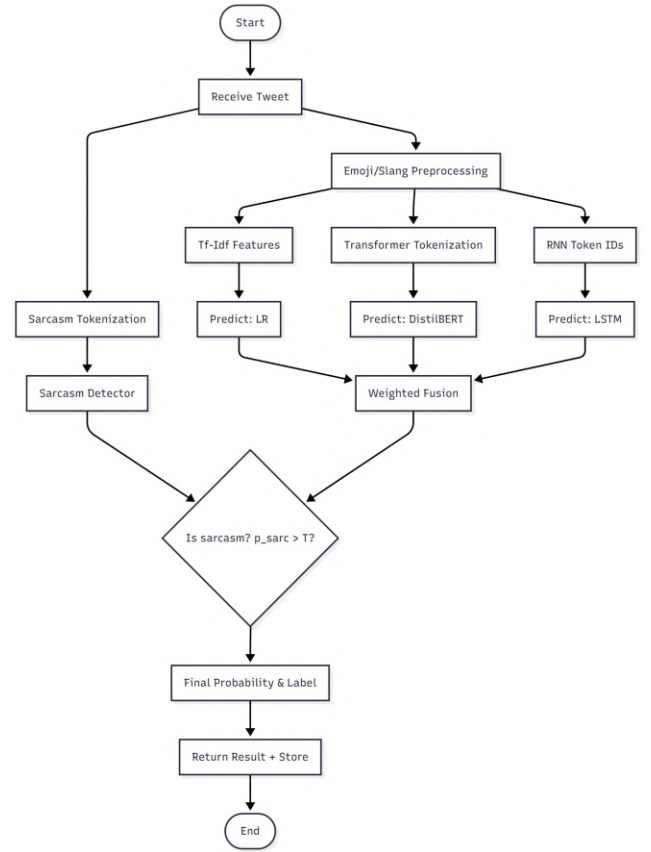


Fig.2 -Processing flowchart showing where sarcasm detection intervenes.

VI. MODEL TRAINING:

The hybrid architecture proposed underwent training in a modular fashion, with each expert model fine-tuned independently, followed by the subsequent ensembling process. For the DistilBERT transformer, fine-tuning followed the procedure of Model Training with 800,000 tweets fine-tuned using the AdamW optimizer at a learning rate of 2×10^{-5} with mini-batches through three epochs, reaching a best validation accuracy of 86.56%. Although the TF-IDF + Logistic Regression classifier and LSTM sequential model were fine-tuned on the entirety of tweets, they instead used 50,000 stratified samples to determine the final parameters, reaching validation accuracies of 78.56% and 74.49%, respectively. Checkpoints of model training were saved at 10,000-step intervals to avoid losing progress or data to memory or power failure, and provide partial updates if retraining was required. While the DistilBERT component of training showed trends of steadily increasing accuracy over the epochs, the validation curve did not similarly improve and eventually levelled out indicating that the model generalized well to the test data, with minimal overfitting.

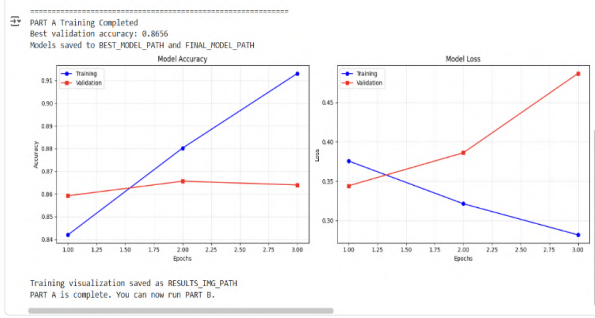


Fig.3- Training and validation accuracy/loss curves for DistilBERT model.

VII. PERFORMANCE EVALUATION:

To evaluate how effective the model has been, we have used various traditional evaluation metrics, namely Accuracy, Precision, Recall, and F1-Score.

These measures assist us in determining how reliable the model is at classifying sentiment by including the percentage of predicted sentiment classes compared to inaccurate sentiment classes.

1. Accuracy

Throughout this work, we wanted to indicate the overall accuracy of the model's predictions and the proportion of tweets classified correctly for the entire sample.

It is expressed as:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (2)$$

In these equations, TP and TN = true positives and true negatives, while FP and FN = false positives and false negatives.

2. Precision

indicating the model's capability to determine all relevant positive cases, thereby defining the model's capability of avoiding false positive across all tweets, is a great reference. It is computed as:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (3)$$

The greater the precision, the greater the credence to predicting a positive sentiment.

3. Recall

Recall can also be referred as "Sensitivity" recall dictates the percentage of actually positive class examples that the model is able to identify. In addition, Recall commonly attempts to reduce slow negatives. defined by:

$$\text{Recall} = \frac{TP}{TP + FN} \quad (4)$$

4. F1-Score

The F1-score introduced here, combines precision and recall into a single score reflecting the average of this two numbers, or a harmonic mean.

It provides a balanced evaluation of accuracy and completeness:

$$F1 = 2 \times \frac{(\text{Precision} \times \text{Recall})}{(\text{Precision} + \text{Recall})} \quad (5)$$

5. Transformer Attention and Prediction Equations:

The self-attention mechanism in the DistilBERT encoder

computes contextual dependencies across tokens using the following formulation:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (6a)$$

where Q , K , and V represent the **query**, **key**, and **value** matrices derived from the input embeddings, and d_k denotes. After attention, layer normalization, and feed-forward processing, the final hidden state corresponding to the $[CLS]$ token, denoted as h_{CLS} , is passed through a dense layer and softmax activation to predict sentiment polarity:

$$P(y | x) = \text{softmax}(W_h h_{CLS} + b_h) \quad (6b)$$

where W_h and b_h are the learned weights and biases of the output classification layer.

6. Ensemble Fusion and Sarcasm Adjustment

The proposed hybrid ensemble combines predictions from the **Transformer**, **Logistic Regression**, and **RNN (BiLSTM)** models using a weighted integration mechanism:

$$P_{\text{final}} = w_t P_{\text{Transformer}} + w_{lr} P_{\text{LR}} + w_{rnn} P_{\text{RNN}} \quad (7a)$$

where w_t , w_{lr} , and w_{rnn} denote the ensemble weights assigned to each model's probability output, satisfying

$$w_t + w_{lr} + w_{rnn} = 1.$$

If the sarcasm detector produces a sarcasm probability p_{sarc} exceeding the threshold T , the final sentiment probability is adjusted as:

$$P'_{\text{final}} = \begin{cases} (1 - P_{\text{final}}), & \text{if strong sarcasm is detected;} \\ P_{\text{final}} \times (1 - \alpha), & \text{if mild sarcasm is detected.} \end{cases} \quad (7b)$$

Here, $\alpha \in [0,1]$ represents the **confidence scaling factor** that proportionally reduces sentiment certainty based on sarcasm intensity.

VIII. RESULTS AND DISCUSSION:

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=== Classification Report: LR ===

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	precision	recall	f1-score	support
0	0.7877	0.7718	0.7797	40000
1	0.7763	0.7921	0.7841	40000
accuracy			0.7819	80000
macro avg	0.7820	0.7819	0.7819	80000
weighted avg	0.7820	0.7819	0.7819	80000

Fig.4- Classification report of Logistic Regression

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=== Classification Report: Transformer ===

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	precision	recall	f1-score	support
0	0.8616	0.8712	0.8664	40000
1	0.8698	0.8600	0.8649	40000
accuracy			0.8656	80000
macro avg	0.8657	0.8656	0.8656	80000
weighted avg	0.8657	0.8656	0.8656	80000

Fig.5- Classification report of DistilBERT Transformer Model

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=== Classification Report: RNN ===

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	precision	recall	f1-score	support
0	0.7225	0.7799	0.7501	40000
1	0.7609	0.7004	0.7294	40000
accuracy			0.7401	80000
macro avg	0.7417	0.7401	0.7397	80000
weighted avg	0.7417	0.7401	0.7397	80000

Fig.6- Classification Report of RNN – LSTM Model

=== Classification Report: Ensemble ===				
	precision	recall	f1-score	support
0	0.8609	0.8726	0.8667	40000
1	0.8709	0.8590	0.8649	40000
accuracy			0.8658	80000
macro avg	0.8659	0.8658	0.8658	80000
weighted avg	0.8659	0.8658	0.8658	80000

Fig.7- Classification report of Hybrid Ensemble Model

Each component model was trained separately to guarantee stable convergence before merging. The DistilBERT model was fine-tuned for three epochs using the AdamW optimizer (learning rate = 2×10^{-5}), reaching a best validation accuracy of 0.8656 (Fig. 5). The Logistic Regression model achieved a validation accuracy of 78.56%, trained on a sample size of 50,000, which established the lexical baseline using TF-IDF features. Finally, the BiLSTM network, which was trained for five epochs on a vocabulary size of 10,002 under the bidirectional configuration, achieved 74.49% validation accuracy which learned dependencies of sequential tokens. (Fig. 6).

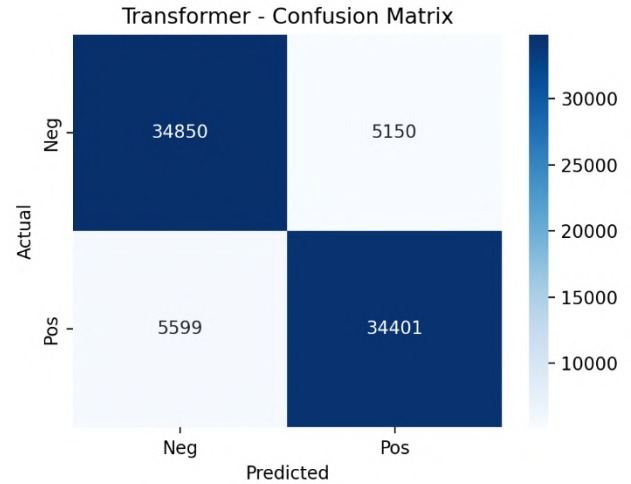


Fig.10- Confusion Matrix of DistilBERT Model

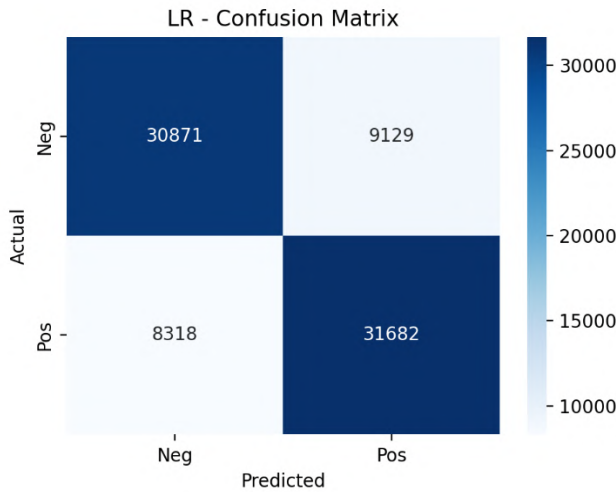


Fig.8- Confusion Matrix of Logistic Regression(TF-IDF)

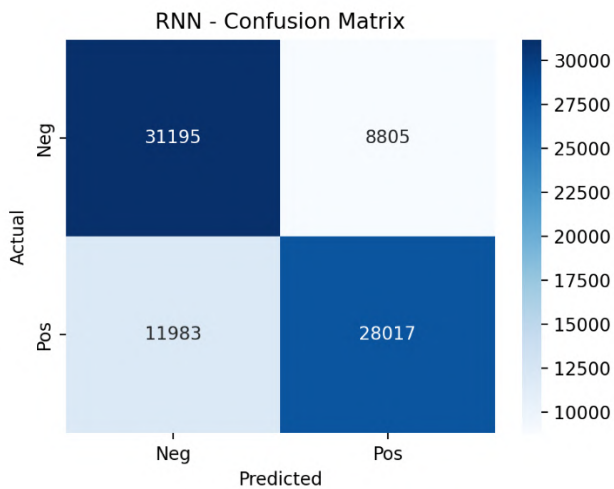


Fig.9- Confusion Matrix of BiLSTM (Sequential RNN)

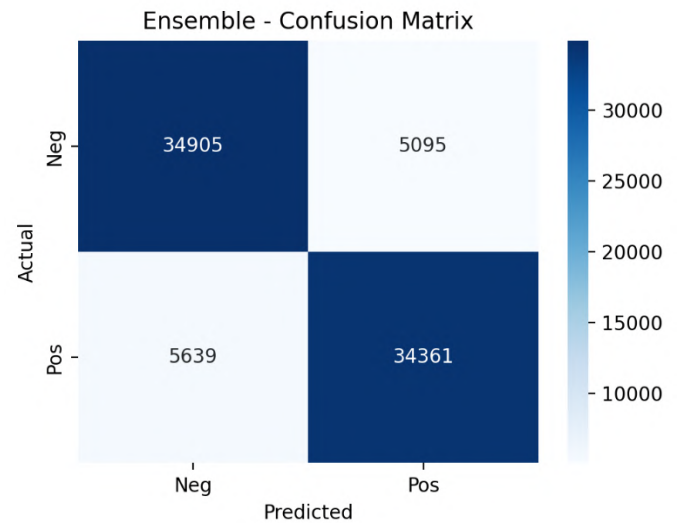


Fig.11- Confusion Matrix of Proposed Hybrid Ensemble Model

The confusion matrices for each sentiment model show how the tweets were classified, by the total number of correctly and incorrectly classified tweets, across both the positive and negative classes, respectively (Figs. 8-11).

The Logistic Regression (TF-IDF) model was moderate, and well-balanced, with 30,871 negative tweets correctly classified and 31,682 positive tweets correctly classified, while incorrect, 9,129 were false positives and 8,318 were false negatives, resulting in overall accuracy of 78.19%.

The hybrid ensemble model increased temporal learning; however, it had a lower accuracy of 74.01%, with 31,195 negative tweets and 28,017 positive tweets correctly classified, but it produced 8,805 false positives and 11,983 false negatives, suggesting it struggles with sarcasm and knowing when sentiment is implicitly reversed.

The DistilBERT transformer model demonstrated increased performance overall and proper contextual understanding, with 34,850 negative tweets and 34,401 positive tweets correctly classified, while having minimal false (5,150 false positives and 5,599 false negatives), and final performance,

was accurate at 86.56% on the testing dataset. Finally, the proposed hybrid ensemble model had the greatest level of performance overall, with 34,905 negative tweets correctly classified and 34,361 positive tweets accurately classified, with only 5,095 false positives and 5,639 false negatives, resulting in final performance, accuracy for the testing dataset at 86.58%. The results indicate that the weighted integration of the ensemble increased performance while reducing the number of errors based on the class imbalance, and increases reliability within the decision making of the model versus each model working independently.

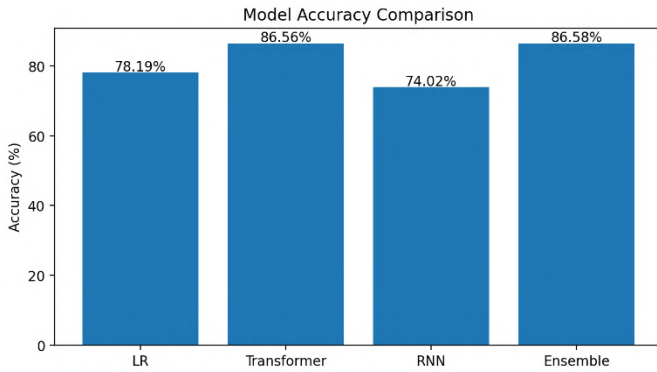


Fig.12- Comparative Model Accuracies For LR, RNN, DistilBERT (Transformer) and Ensemble Model

The Transformer and Ensemble Perform better than traditional architectures, as shown by the comparative accuracy chart(Fig.12).

By combining Transformer-based semantic understanding, BiLSTM sequential context, and TF-IDF lexical features, the multi-level feature fusion approach improves stability and adaptability.

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HYBRID SENTIMENT ANALYSIS - INTERACTIVE MODE
=====
All 4 models are loaded and ready.
Enter a sentence to analyze. Type 'quit' or 'exit' to stop.
=====

Enter text: I am so happy with the customer service, they were amazing.

MODEL BREAKDOWN:
- Sarcasm      : Not Sarcastic (Conf: 0.26)
- Transformer  : Positive (Prob Positive: 1.00)
- Logistic Reg. : Positive (Prob Positive: 0.85)
- RNN (LSTM)   : Positive (Prob Positive: 0.96)

FINAL PREDICTION:
> Weighted Sentiment (Before Sarcasm): 0.96

>> FINAL SENTIMENT: Positive
>> CONFIDENCE: 0.961
=====

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Fig.13- Example positive tweet prediction

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Enter text: I am extremely frustrated and disappointed.

MODEL BREAKDOWN:
- Sarcasm      : Not Sarcastic (Conf: 0.83)
- Transformer  : Negative (Prob Positive: 0.00)
- Logistic Reg. : Negative (Prob Positive: 0.14)
- RNN (LSTM)   : Negative (Prob Positive: 0.06)

FINAL PREDICTION:
> Weighted Sentiment (Before Sarcasm): 0.04

>> FINAL SENTIMENT: Negative
>> CONFIDENCE: 0.958
=====

Enter text: Oh great, my flight is delayed again. Just wonderful.

MODEL BREAKDOWN:
- Sarcasm      : Sarcastic (Conf: 0.99)
- Transformer  : Negative (Prob Positive: 0.04)
- Logistic Reg. : Positive (Prob Positive: 0.63)
- RNN (LSTM)   : Positive (Prob Positive: 0.94)

FINAL PREDICTION:
> Weighted Sentiment (Before Sarcasm): 0.34

>> FINAL SENTIMENT: Negative
>> CONFIDENCE: 0.661
=====

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Fig.14- Negative sentiment classification

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=====
Enter text: "Amazing" service. They hung up on me twice.

MODEL BREAKDOWN:
- Sarcasm      : Sarcastic (Conf: 0.91)
- Transformer  : Positive (Prob Positive: 0.73)
- Logistic Reg. : Positive (Prob Positive: 0.52)
- RNN (LSTM)   : Positive (Prob Positive: 0.81)

FINAL PREDICTION:
> Weighted Sentiment (Before Sarcasm): 0.71
> SARCASM DETECTED! Adjusting sentiment.

>> FINAL SENTIMENT: Negative
>> CONFIDENCE: 0.707
=====

Enter text: I just love being told I have to redo my entire project.

MODEL BREAKDOWN:
- Sarcasm      : Sarcastic (Conf: 0.98)
- Transformer  : Negative (Prob Positive: 0.06)
- Logistic Reg. : Positive (Prob Positive: 0.66)
- RNN (LSTM)   : Positive (Prob Positive: 0.69)

FINAL PREDICTION:
> Weighted Sentiment (Before Sarcasm): 0.30

>> FINAL SENTIMENT: Negative
>> CONFIDENCE: 0.695
=====

```

Fig.15- Sarcasm detection and Sentiment Inversion

To process unseen tweet inputs in real time, an interactive inference mode was put into place. To generate the final sentiment output with a corresponding confidence score, the model combines predictions from four separate modules: DistilBERT, Logistic Regression (TF-IDF), BiLSTM (Sequential RNN), and a RoBERTa-based Sarcasm Detector. Every test case shows how the model can recognise sarcasm, incorporate predictions from multiple models, and modify sentiment orientation appropriately. When a text is marked as sarcastic with a confidence level higher than 0.90, In order to improve interpretive reliability for irony-laden expressions, the system reverses the sentiment polarity using Equation (7).

IX. CONCLUSION:

DistilBERT, Logistic Regression, and BiLSTM models are all integrated into a single weighted ensemble architecture in this study's hybrid sarcasm-aware sentiment analysis framework. By skilfully combining lexical, contextual, and sequential representations, the suggested system makes it possible to fully comprehend the implicit and explicit emotions conveyed in social media text. In the presence of ironic or unclear content, a specialised sarcasm detection module based on RoBERTa dynamically recalibrates sentiment polarity, further improving interpretability. The ensemble model outperformed all individual submodels with an overall accuracy of 86.58%, according to experimental evaluation on the Sentiment140 dataset. The confusion matrix's findings verified that both the

positive and negative sentiment classes had balanced classification performance.

Furthermore, the system's resilience in managing intricate linguistic elements like emojis, slang, and sarcasm was confirmed by qualitative analysis using case studies. All things considered, the suggested architecture creates a sentiment analysis solution that is scalable, interpretable, and context-aware and appropriate for real-time social media applications. To improve its adaptability across a variety of contexts, future studies will concentrate on expanding this framework to multilingual sentiment detection and fine-grained emotion classification.

X. ACKNOWLEDGMENT

We sincerely thank our institution and our faculty for their guidance and support throughout this project. Additionally, we acknowledge the open-source community for providing essential datasets, tools, and frameworks that helped our implementation.

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